

Quadratic Sequences

AQA · KS3 Year 7–8

Answer each question in your workbook. Show your working.

Method

First differences Second differences (constant?) $a = \text{second difference} \div 2$ Subtract an^2 find $bn + c$

Combine: $an^2 + bn + c$

Common mistakes

- Forgetting to halve the second difference.
- Slips subtracting the an^2 values.
- Not finding the linear part correctly.
- Sign errors with a negative linear part.

Worked example

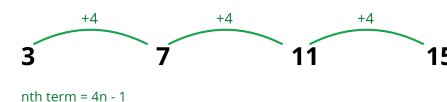
Find nth term of 3, 8, 15, 24, 35.

Second difference 2 $a = 1$ (n^2). Subtract n^2 : 2, 4, 6, 8, 10 = $2n$.

nth term = $n^2 + 2n$.

Visual aid

Visual Example



A quadratic sequence has a constant SECOND difference. Halving it gives the coefficient of n^2 ; subtracting the an^2 values leaves a simpler linear sequence to finish the rule.

Questions

Answer each question in your workbook. Show your working.

Q1 of 8

[2] ● ○ ○ ○ ○ ○

Find first and second differences: a) 2, 5, 10, 17, 26 b) 4, 7, 12, 19, 28

Q2 of 8

[2] ● ○ ○ ○ ○ ○

Quadratic? Show second differences: a) 1, 4, 9, 16 b) 2, 4, 6, 8

Q3 of 8

[3] ● ● ○ ○ ○ ○

Find the nth term: a) 2, 5, 10, 17, 26 b) 4, 7, 12, 19, 28 c) 3, 9, 19, 33, 51

Q4 of 8

[3] ● ● ○ ○ ○ ○

nth term = $n^2 + n$. a) 1st term b) 5th term c) 10th term

Q5 of 8

[3] ● ● ● ○ ○ ○

Find the nth term: a) 0, 3, 8, 15, 24 b) 5, 14, 27, 44, 65 c) 3, 8, 15, 24

Q6 of 8

[3] ● ● ● ○ ○ ○

Tile pattern: 2, 6, 12, 20, ... a) Tiles in pattern 5. b) nth term.

Q7 of 8

[4] ● ● ● ● ○ ○

Sequence 1, 6, 15, 28, 45. a) Show second difference constant. b) nth term. c) 10th term.

Q8 of 8

[5] ● ● ● ● ● ○

nth term = $n^2 - 3n + 2$. a) First four terms. b) Show 2nd difference is 2. c) Which term = 30?